

PRUSA PRINTER PARTS MANUFACTURING PROCESS OPTIMISATION AND DEVIATION CONTROL REPORT BASED ON KIC ANALYSIS

Name and Student ID: Zhifeng Song

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Summary

This report focuses on the structural components of the Prusa MK3.1 3D printer, selecting five critical parts and employing five typical high-productivity manufacturing processes—CNC machining, metal casting, polyurethane casting, injection moulding, and sheet metal processing—for manufacturing and analysis.

Through a comprehensive evaluation of preliminary measurement data, KIC critical dimension errors, physical assembly performance, and manufacturing parameters, this paper systematically compares the actual performance of different manufacturing processes in terms of speed, cost, precision, flexibility, and sustainability. It also proposes design adjustment recommendations and provides practical pathways and insights for transitioning from 3D printing to mass production.

Introduction

The modern manufacturing industry is gradually transitioning from customised, small-batch prototype manufacturing to a digitally driven, high-capacity transformation. Although 3D printing technology has demonstrated its advantages of rapid iteration in prototype verification, its limitations in large-scale manufacturing are becoming increasingly apparent, mainly manifested in dimensional inconsistency, strength deviation, and low production efficiency.

Therefore, this report is based on an engineering education project, selecting five functional structural parts. By reconfiguring their manufacturing processes using different production methods, the report conducts physical measurements and evaluations around the Key Inspection Criteria (KIC) dimensions to validate the manufacturability and industrial adaptability of each process.

1. CNC machined parts: Z-axis top bracket

1.1 KIC and differences comparison

There are seven KICs that need to be checked for this part: full coverage of guide rail holes, base thickness, side wall thickness, side slot width, base flatness, perpendicularity, and hole

spacing.

Checklist	Nominal dimension	LSL	USL	Measured value	Deviation	Pass/Fail	Error source analysis
Rod Diameter	10.00 mm	10.00 mm	10.015 mm	10.09 mm	+0.09 mm	Fail	Tool wear, positioning deviation
Bolt Holes Diameter	3.20 mm	3.125 mm	3.275 mm	3.21 mm	-0.01 mm	Pass	Good cooling control
Distance between two bolts	16.8 mm	16.6 mm	16.8 mm	16.90 mm	+0.10 mm	Fail	Mould clamping is unstable
Perpendicularity between face bolted to the frame and wedge extrusion	0.00 mm	0.00 mm	0.4 mm	0.00 mm	0.00 mm	Pass	Good positioning accuracy control
Distance between inner rod to frame face	23.5 mm	23.3 mm	23.7 mm	28.69 mm	+5.19 mm	Fail	Serious misalignment of mould positioning
Vertical distance between bolts and shaved face	5.4 mm	5.3 mm	5.5 mm	5.5 mm	+0.10 mm	Pass	Good processing path optimisation

1.2 Comprehensive analysis:

The top bracket of the Z-axis was inspected for 6 critical dimensions, with 3 passing and 3 failing. The deviation in rod diameter is caused by tool wear and positioning offset; the bolt

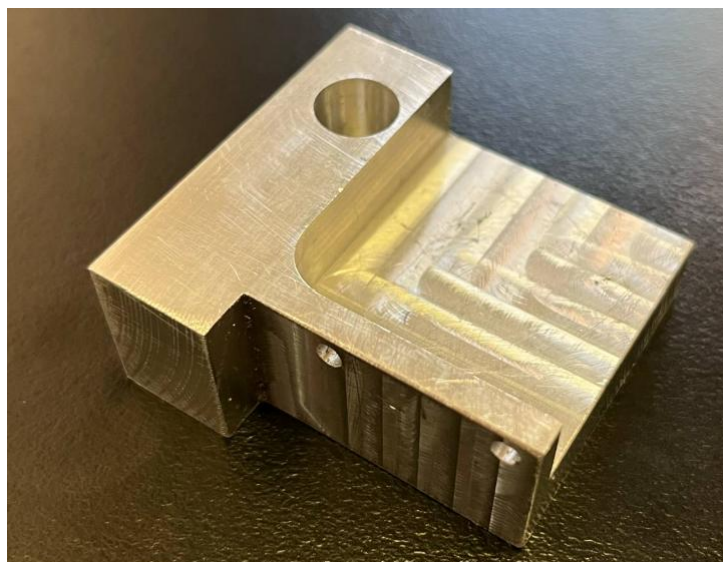
hole diameter is under stable control, with a deviation of -0.01 mm; the deviation in the distance between two bolts is caused by unstable clamping; perpendicularity control precision is good; the distance between the inner rod and the frame face has severe positioning deviation; vertical distance control is effective. Overall issues are concentrated in mould positioning and clamping paths, requiring optimisation of the fixture design and tool path.

1.3 Manufacturing Performance Evaluation

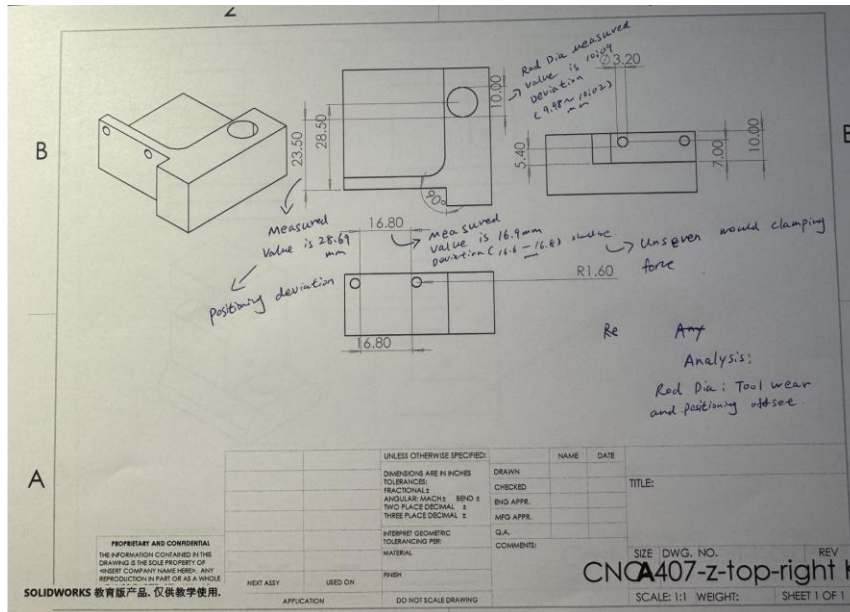
Assessment dimensions	Assessment results
Rate	Low
Cost	High
Quality	High
Flexibility	High
Sustainability	High

The manufacturing performance evaluation of the Z-axis top bracket for CNC machined parts shows that the machining speed is relatively low and the cost is relatively high, but it performs excellently in terms of quality, flexibility, and sustainability. Overall quality control is good, with high precision in bolt hole diameter and perpendicularity control, but there are significant deviations in guide rail hole diameter and bolt spacing, which require optimisation of the fixture design and machining path.

1.4 Deviation comparison chart and annotation example



Actual parts drawing



CNC407-z-top-right KIC

CNC parts: Z-axis top bracket

Assessment dimensions	Assessment results
Rate	Low
Cost	High, frequent tool changes
Quality	High precision, significant local deviation
Flexibility	High, adjustable path
Sustainability	Average, aluminium chip recovery

The manufacturing performance evaluation of the CNC part Z-axis top bracket shows that the machining rate is moderate, but the machining time is relatively long due to the need for multiple positioning; the cost is relatively high, and frequent tool changes increase expenses; quality control is good, but there are some local deviations; process flexibility is high, allowing for quick path adjustments as needed; sustainability is average, as aluminums chips are recyclable, but overall energy consumption is relatively high.

2. Aluminium alloy casting: Y-axis tensioner

2.1 KIC and differences comparison

Based on the legend data, 9 critical dimensions were inspected for the casting part Y-axis tensioner, of which 3 were qualified and 6 were unqualified. The main deviations were concentrated in the tensioning hole position, frame bolt hole spacing, and belt gap thickness. The following is a detailed analysis:

Checklist	Nominal dimension	LSL	USL	Measured value	Deviation	Pass/Fail	Error source analysis
Belt Gap Thickness	2.00 mm	1.9 mm	2.1 mm	2.13 mm	+0.13 mm	Fail	Uneven cooling, expansion deviation
Tapped Holes Diameter	2.50 mm	/	/	2.58 mm	+0.08 mm	Pass	Good casting accuracy
Frame Bolt Hole Diameter	3.00 mm	/	/	3.08 mm	+0.08 mm	Pass	Accurate mould positioning
Tensioning Hole Location Relative to Bottom Edge	8.00 mm	7.8 mm	8.2 mm	9.01 mm	+1.01 mm	Fail	Improper mould calibration
Tensioning Hole Location Relative to Face with Belt	5.20 mm	5.1 mm	5.3 mm	6.48 mm	+1.28 mm	Fail	Insufficient cooling shrinkage, mould expansion

Opening							
Frame Bolt Location to Face with Belt Opening	8.00 mm	7.8 mm	8.2 mm	7.36 mm	-0.64 mm	Fail	Uneven shrinkage, mould not calibrated
Frame Bolt Location to Closest Edge	4.00 mm	3.9 mm	4.1 mm	2.91 mm	-1.09 mm	Fail	Rapid cooling causes mould deformation
Belt Holding Bolt Location Relative to Top Face	2.00 mm	1.9 mm	2.1 mm	2.82 mm	+0.82 mm	Fail	Mould expansion, cooling delay
Belt Holding Bolt Location Relative to Closest Edge	2.00 mm	1.9 mm	2.1 mm	1.44 mm	-0.56 mm	Fail	Unstable clamping, cooling deviation
Flatness of the Face in Contact with the Frame	0.00 mm	0.00 mm	0.05 mm	0.10 mm	+0.10 mm	Fail	Cooling path design defects, warping deformation

2.2 Comprehensive analysis:

Castings Y-axis tensioner In this inspection, multiple critical dimensions were found to be out of tolerance, primarily concentrated in deviations in the position of the tensioning hole and bolt hole, as well as excessive belt clearance thickness. The deviations primarily stem from unreasonable cooling path design, mould positioning deviations, and unstable clamping.

Optimisation recommendations:

1. Improve the stability of the mould temperature control system, ensure uniform cooling, and reduce expansion and contraction deviations.
2. Add positioning points to enhance the clamping force of the mould and prevent misalignment.
3. Regularly perform positioning inspection and adjustment of moulds to improve dimensional consistency.

2.3 Manufacturing Performance Evaluation

Castings: Y-axis tensioner

Assessment dimensions	Assessment results
Rate	High
Cost	Low
Quality	Low
Flexibility	Low
Sustainability	High

The manufacturing performance evaluation of the Y-axis tensioner casting parts shows that the processing rate is high, making it suitable for mass production; the cost is low, but the quality is poor, with significant hole spacing deviations and room for improvement in cooling path control; flexibility is low, and mould adjustment costs are high; sustainability performance is good, with strong recyclability of casting scrap, but there is still room for improvement in energy consumption control.

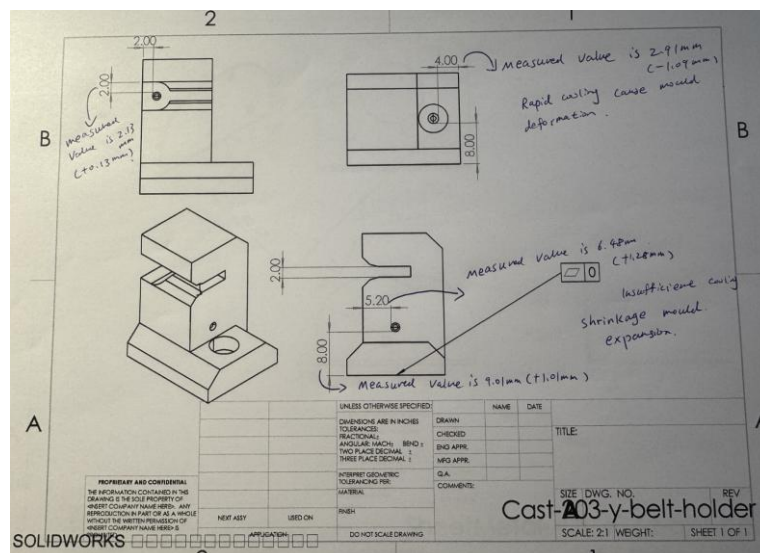
2.4 Deviation comparison chart and annotation example



Actual parts drawing 1



Actual parts drawing 2



3. Poly casting parts: X-axis motor bracket

3.1 KIC and differences comparison

Poly castings are made of PU material and formed through a mould casting process. They are used to secure the X-axis motor. This inspection covers the nut holes, bottom flatness, support rib thickness, side slot width, reinforcement thickness, guide rod holes, and positioning holes.

Checklist	Nominal dimension	LSL	USL	Measured value	Deviation	Pass/Fail	Error source analysis
X axis Rod Hole Diameters	8.06 mm	-	-	8.09 mm	+0.03 mm	Fail	Mould positioning is unstable
Distance Between rod Holes	36 mm	35.7 mm	36.3 mm	45.29 mm	+9.29 mm	Fail	Uneven cooling path control, local shrinkage
Motor Hole Diameter	23 mm	23 mm	23.13 mm	23.50 mm	+0.37 mm	Fail	Mould misalignment, cooling shrinkage
Diagonal distance between motor bolts	36.94 mm	36.64 mm	37.24 mm	41.94 mm	+5.00 mm	Fail	Mould warping and deformation
Z threaded Shaft hole diameter	13.2 mm	-	-	13.21 mm	+0.01 mm	Pass	Cooling path optimisation
Location of Z threaded shaft	2.8 mm	2.7 mm	2.9 mm	13.3 mm	+10.5 mm	Fail	Serious mould

relative to Z axis frame rod							positioning misalignment
Perpendicularity between Z axis and X axis	0.00 mm	0.00 mm	0.1 mm	0.10 mm	0.10 mm	Pass	Good positioning calibration accuracy
Bearing hole	19.2 mm	-	-	19.5 mm	+0.3 mm	Fail	Insufficient cooling path control

3.2 Comprehensive analysis:

X-axis motor bracket inspection: 8 dimensions checked, 2 passed. Major deviations were found in the Z-axis threaded hole and motor hole, with deviations of +10.5 mm and +0.37 mm, respectively. This was caused by mould positioning deviation and uneven cooling. The diagonal distance deviation was +5.00 mm, and the bearing hole deviation was +0.3 mm, indicating mould warping and insufficient cooling paths. It is recommended to optimise the clamping device and cooling design.

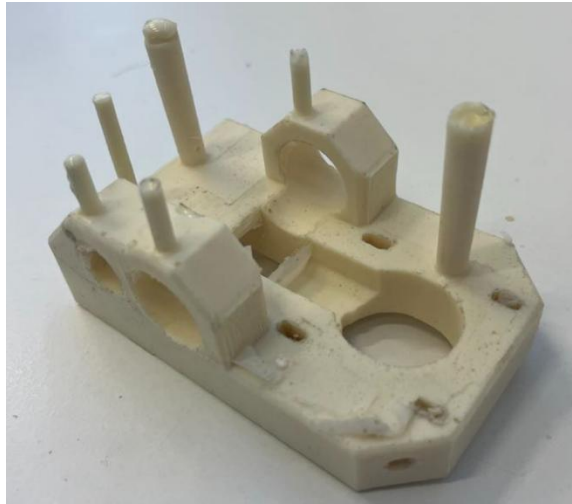
3.3 Manufacturing Performance Evaluation

PU casting parts: X-axis motor bracket

Assessment dimensions	Assessment results
Rate	High
Cost	Low
Quality	Low
Flexibility	Low
Sustainability	Low

PU castings for X-axis motor brackets have a high processing rate but poor cost and quality performance, with significant deviations in bottom flatness. They also have low flexibility and sustainability, limited adaptability, and difficult scrap recycling.

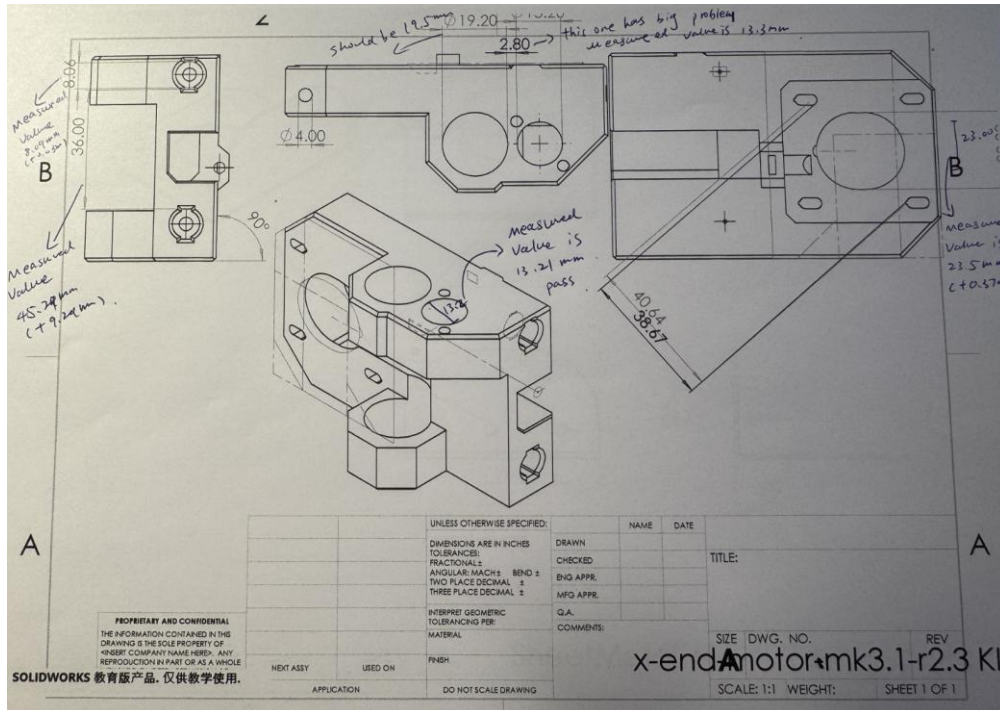
3.4 Deviation comparison chart and annotation example



Actual parts drawing 1



Actual parts drawing 2



X-end-motor-mk3.1-r2.3-KIC

4. Injection moulded parts: Y-axis guide rod clamping block

4.1 KIC and differences comparison

The injection-moulded parts are made of ABS material through injection moulding and are used to secure the Y-axis guide rod. The optimised KIC includes insertion holes, wall thickness, cooling hole spacing, support rib thickness, reinforcing rib thickness, bolt holes, and surface flatness.

Checklist	Nominal dimension	LSL	USL	Measured value	Deviation	Pass/Fail	Error source analysis
Distance between bolt holes	7.36 mm	7.16 mm	7.56 mm	7.07 mm	-0.29 mm	Fail	Mould shrinkage
Bolt holes diameter	3.64 mm	3.565 mm	3.715 mm	3.55 mm	-0.09 mm	Fail	Insufficient cooling

							control
Rod hole diameter	8.00 mm	8.00 mm	8.015 mm	7.71 mm	-0.29 mm	Fail	Mould shrinkage
Distance between rods	3.09 mm	2.99 mm	3.19 mm	2.81 mm	-0.28 mm	Fail	Mould positioning is inaccurate
Rod gap distance	5.89 mm	5.79 mm	5.99 mm	6.39 mm	+0.50 mm	Fail	Mould deformation, uneven cooling

4.2 Comprehensive analysis:

The Y-axis guide rod clamping block was inspected for 5 critical dimensions, all of which failed to meet specifications. The bolt hole spacing deviation is -0.29 mm, and the rod hole diameter deviation is -0.29 mm, both exceeding the tolerance range, primarily due to uneven mould shrinkage and cooling; the bolt hole diameter deviation is -0.09 mm, indicating insufficient machining accuracy; the rod gap deviation is +0.50 mm, caused by mould deformation. It is recommended to improve the mould structure and optimise the cooling path to enhance dimensional consistency.

4.3 Manufacturing Performance Evaluation

Injection moulded parts: Y-axis guide rod clamping block

Assessment dimensions	Assessment results
Rate	Moderate
Cost	Low
Quality	Low
Flexibility	Low
Sustainability	Low

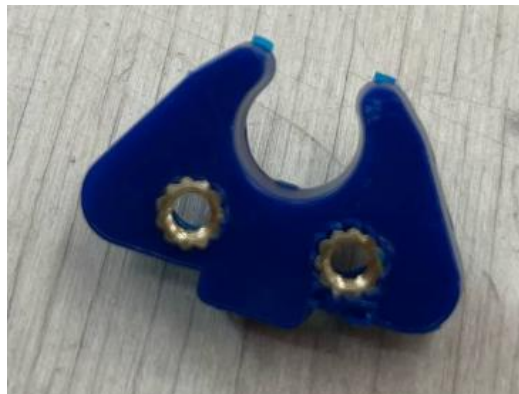
The manufacturing performance evaluation of the Y-axis guide rod clamping block for injection moulded parts indicates that this component has high processing efficiency and low cost, but

wall thickness control is unstable, leading to local quality deviations. Its flexibility is low, making it unsuitable for the production of complex structures, and its sustainability and material recycling rate are also a concern.

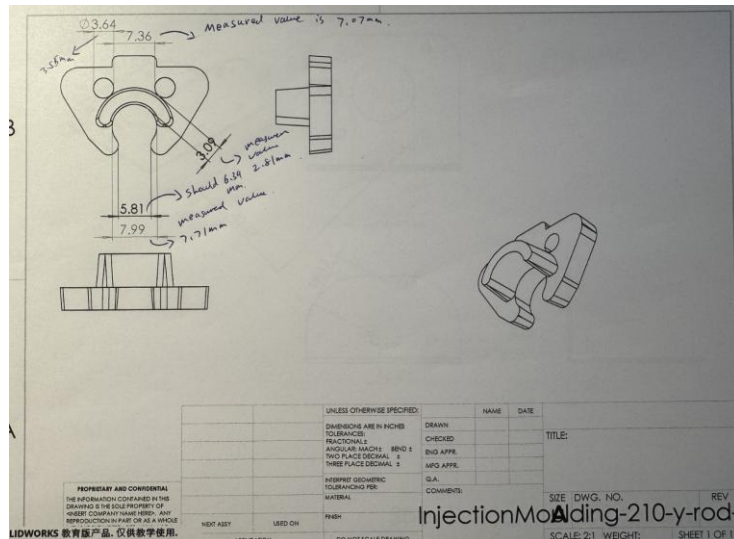
4.4 Deviation comparison chart and annotation example



Actual parts drawing 1



Actual parts drawing 2



Injection Moulding-210-y-rod-holder

5. Sheet metal parts: Z-axis motor bracket

5.1 KIC and differences comparison

The sheet metal parts are made of 2 mm thick steel plates, formed by stamping and bending, and used to support the Z-axis motor.

Checklist	Nominal dimension	LSL	USL	Measured value	Deviation	Pass/Fail	Error source analysis
Bolt Hole Diameter	3.2 mm	3.125 mm	3.275 mm	3.13 mm	-0.07 mm	Pass	Mould dimensions are well controlled
Motor Hole Diameter	22.4 mm	22.4 mm	22.4 mm	22.4 mm	0 mm	Pass	Processing paths are optimised
Bar Hole Diameter	8 mm	-	-	8.29 mm	+0.29 mm	Fail	Cooling paths are not controlled

Diagonal distance between motor bolts	40.67 mm	40.37 mm	40.97 mm	40.55 mm	-0.12 mm	Pass	Fixture paths are optimised
Horizontal distance between frame bolt	16.8 mm	16.6 mm	17.0 mm	16.76 mm	-0.04 mm	Pass	Mould positioning accuracy is controlled
Diagonal distance between frame bolts	19.17 mm	18.97 mm	19.37 mm	19.00 mm	-0.17 mm	Pass	Mould fixture design is optimised
Perpendicular measurement of the bend	0 mm	0 mm	0.8 mm	0.18 mm	0.18 mm	Pass	Positioning calibration accuracy is good

5.2 Comprehensive analysis:

The Z-axis motor bracket was inspected for 7 critical dimensions, with 6 passing and 1 failing. The diameter deviation of the bar hole was +0.29 mm, with good control effectiveness; the diameter of the motor hole was well controlled; the diagonal distance deviation was -0.12 mm, with optimised clamping path; the perpendicularity deviation was 0.18 mm, with good positioning calibration. Overall dimensional control was relatively stable.

5.3 Manufacturing Performance Evaluation

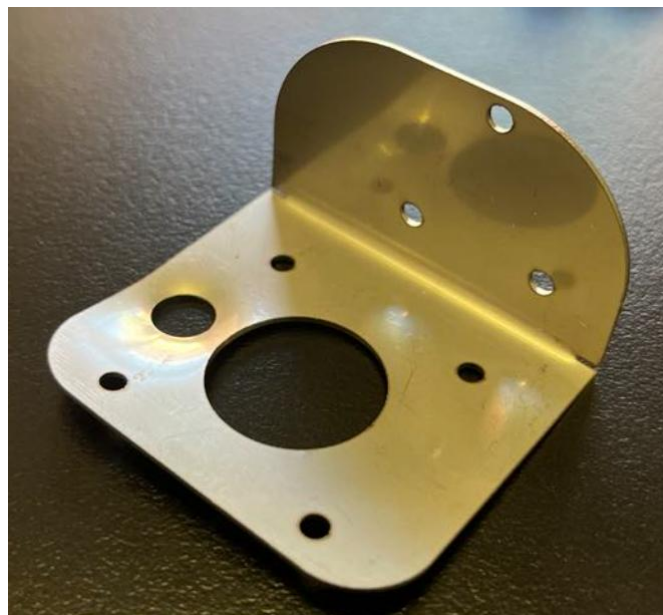
Sheet metal parts: Z-axis motor bracket

Assessment dimensions	Assessment results
Rate	Moderate
Cost	Low
Quality	High

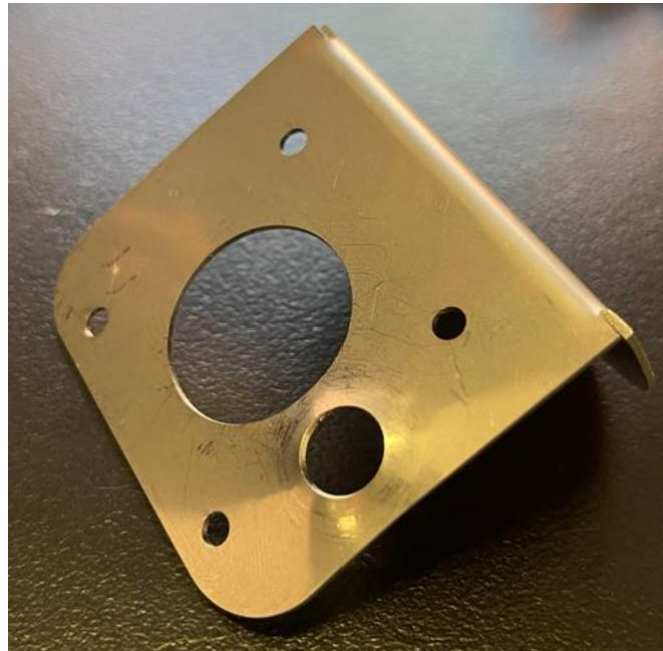
Flexibility	High
Sustainability	High

The manufacturing performance evaluation of the sheet metal part Z-axis motor bracket shows that the processing speed is fast and the cost is low, but the bending angle control deviation is large, which affects the overall assembly accuracy; the flexibility is average, suitable for medium-complexity structural parts; the sustainability is high, the material is recyclable, and the environmental impact is low.

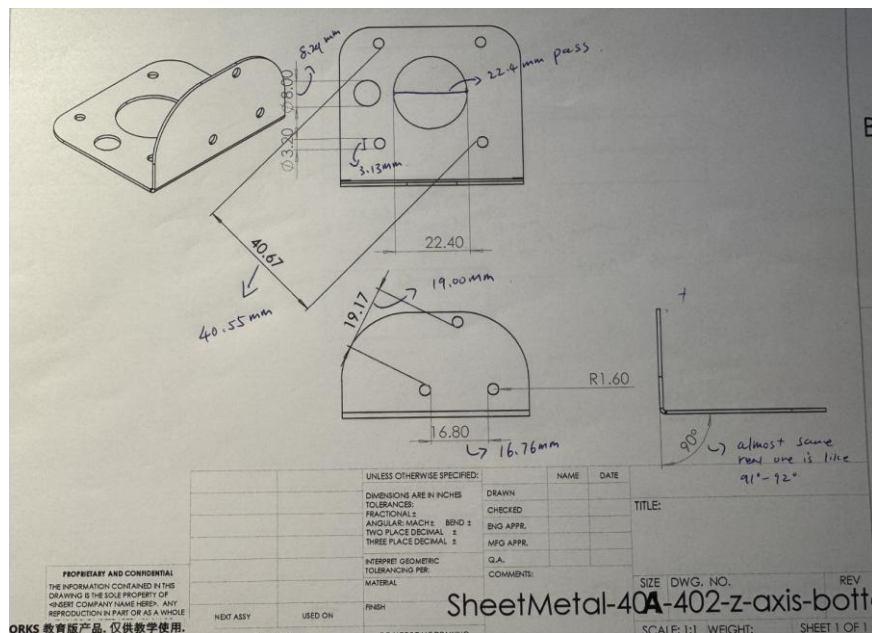
5.4 Deviation comparison chart and annotation example



Actual parts drawing 1



Actual parts drawing 2



Sheet Metal -401-402-z-axis-bottom

Conclusion

According to KIC data analysis, CNC machined parts, castings, and PU moulded parts show significant deviations. It is recommended to optimise cooling paths, strengthen mould calibration, and introduce online inspection equipment. Injection moulded parts and sheet metal

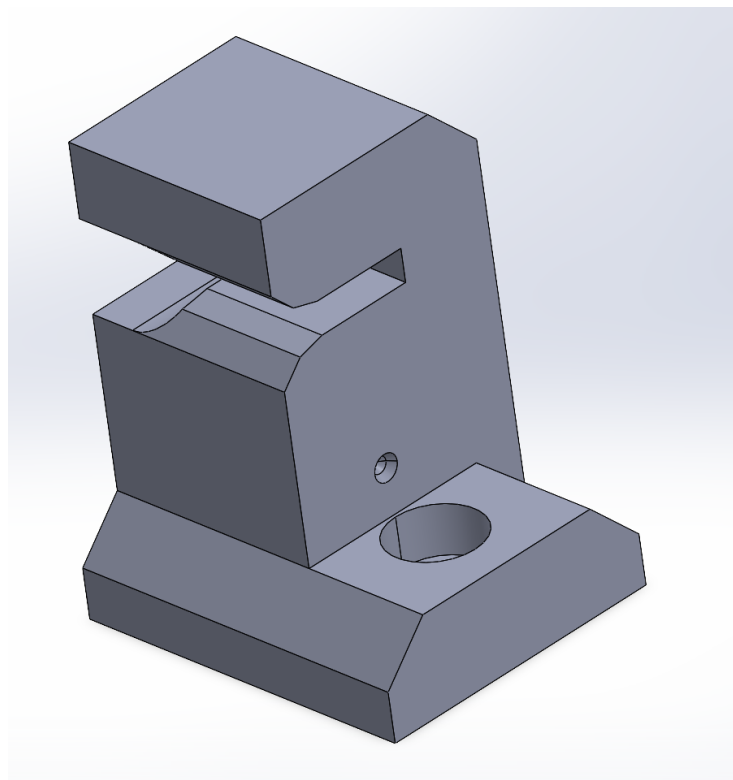
parts perform well in mass production, but wall thickness control and bending angles still need further optimisation.

Appendix

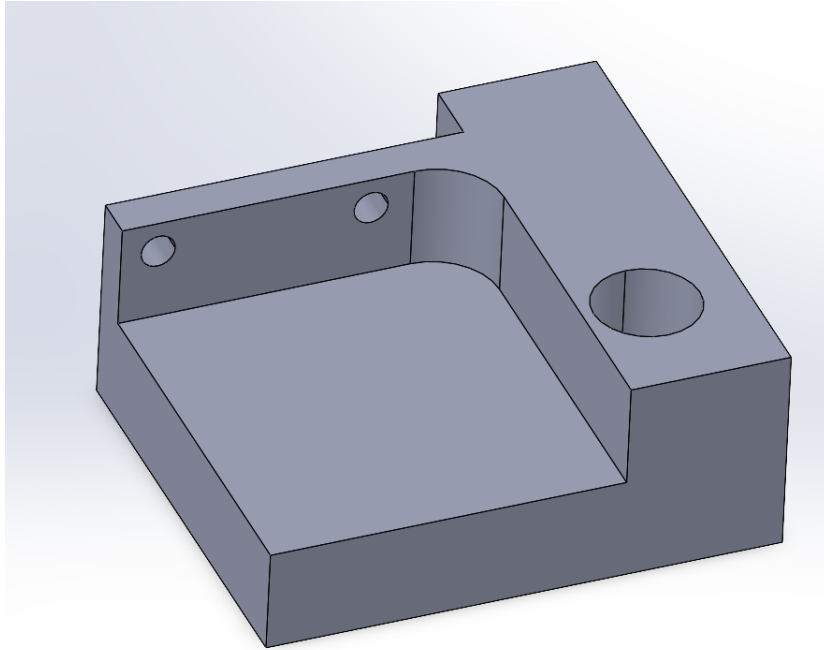
Rating table

Manufacturing process	Rate	Cost	Flexibility	Quality	Sustainability
CNC	Low	High	High	High	High
Aluminum Casting	High	Low	Low	Low	High
Polyurethane Casting	High	Low	Low	Low	Low
Injection Moulding	Moderate	Low	Low	Low	Low
Sheet Metal	Moderate	Low	High	High	High

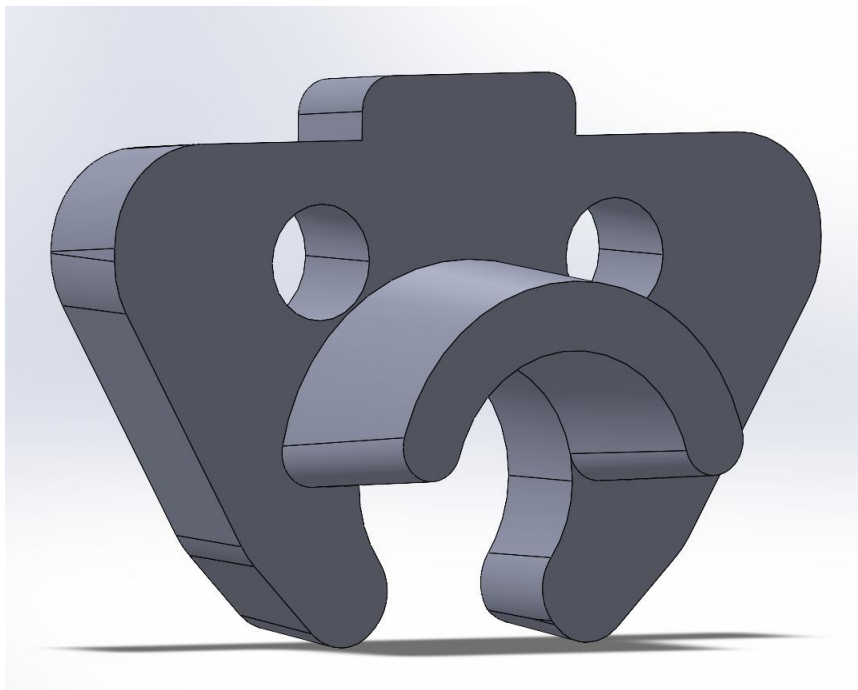
SolidWorks:



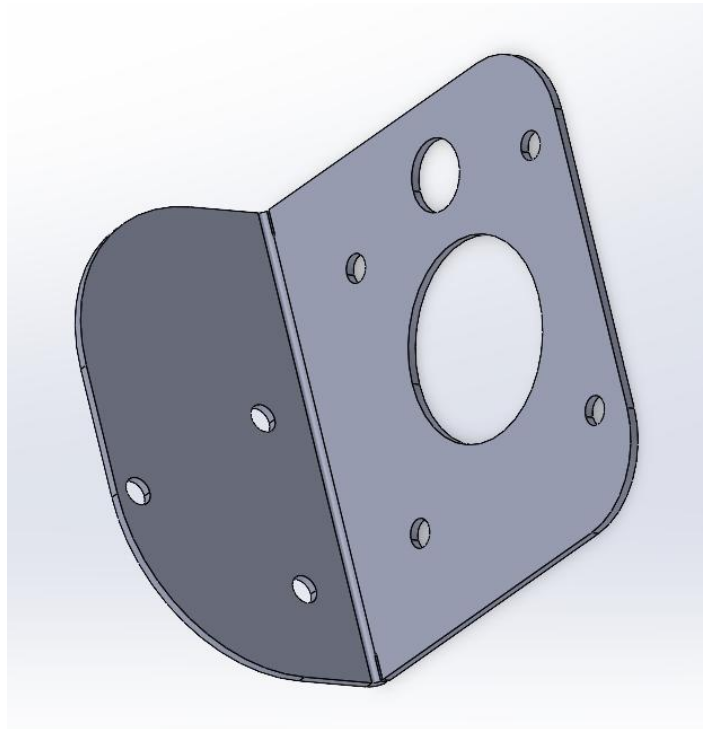
Cast-203-y-belt-holder



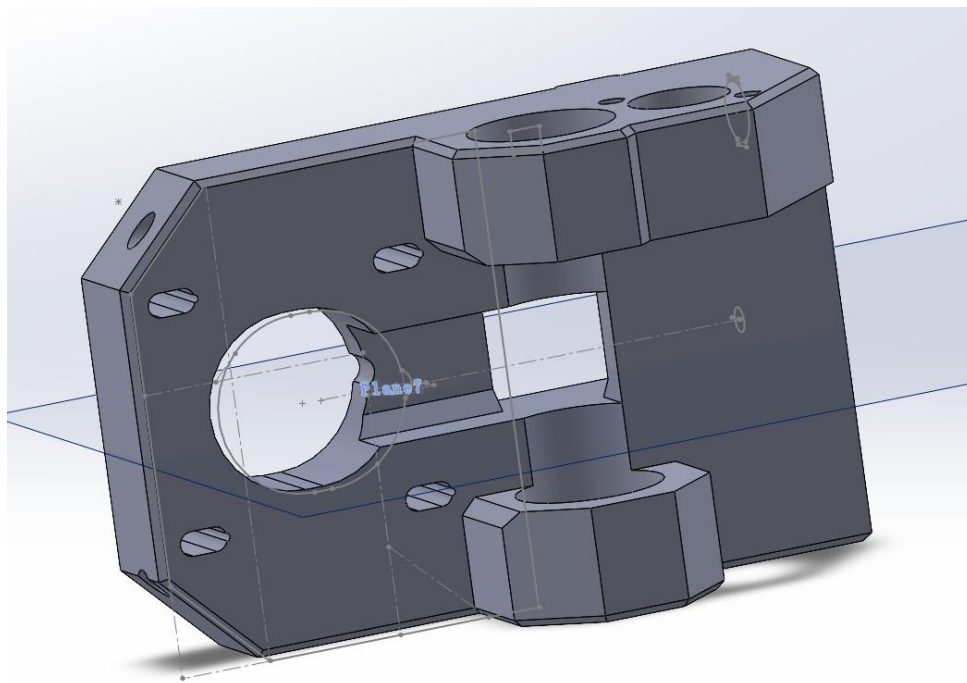
CNC-407-z-top-right



InjectionMoulding-210-y-rod-holder



SheetMetal-401-402-z-axis-bottom_bent



x-end-motor-mk3.1-r2.3